20 **C** is a circle with centre the origin.

L is the straight line with equation $x + 2y = 5$

C and **L** intersect at the points A and B.
The coordinates of A are $(-7, 6)$

Work out the coordinates of B.
You must show all your working.

Radius squared = $(-7)^2 + 6^2 = 85$

1 mark
Centre is the origin.

Circle equation: $x^2 + y^2 = 85$

Line gives $x = 5 - 2y$

1 mark
Substitute the line into the circle.

$(5 - 2y)^2 + y^2 = 85$

$25 - 20y + 4y^2 + y^2 = 85$

$5y^2 - 20y - 60 = 0$

1 mark
Collect all terms.

$y^2 - 4y - 12 = 0$

$(y - 6)(y + 2) = 0$

1 mark
Factorise to find both intersections.

A has $y = 6$. For B, $y = -2$.

$x = 5 - 2(-2) = 9$, so B = $(9, -2)$.

1 mark
Choose the other point and recover x.

(9 , −2)

(Total for Question 20 is 5 marks)

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